

1 Entropy, relative entropy, and mutual information

Reading: Chapter 2 of Cover and Thomas

Review

- Overview of course
- Probability review

Preview of Lec. 3–6

- Entropy (H): a measure of information, defined for a probability distribution P_X
- Mutual Information (I): a measure of the amount of information one random variable contains about the other, defined for two or more random variables, e.g., $X, Y \sim P_{X,Y}$
- Relative Entropy (D): a measure of the distance between two probability distributions, e.g., P, Q

1.1 Entropy

Question 1. *How much information (surprise) we get when we observe a random event E ?*

For the event E , we want to define a measure of information $\tilde{H}(E) \in \mathbb{R}^+$ that satisfies the following properties:

- 1) Should be a function of $\Pr(E)$
- 2) Continuous in $\Pr(E)$
- 3) If $\Pr(E) \uparrow$, then $\tilde{H}(E) \downarrow$ (rare events are more surprising)
- 4) If $E_1 \perp E_2$ (E_1 and E_2 are independent), then $\tilde{H}(E_1 \cap E_2) = \tilde{H}(E_1) + \tilde{H}(E_2)$

Definition 1.

$$\tilde{H}(E) \triangleq -\log_2 \Pr(E) \text{ (bits)} \quad (1)$$

Remarks:

- 1) – sign for property 3)
- 2) $-\log_2 \Pr(E)$ (bits) vs. $-\ln \Pr(E)$ (nats), $1 \text{ nat} \approx 1.44 \text{ bits}$
- 3) “bits” – base of all the digital systems

Question 2. *What is the **average** amount of information one obtains by observing whether or not event E occurs?*

The average amount of information is

$$\Pr(E)\tilde{H}(E) + \Pr(E^c)\tilde{H}(E^c).$$

Think about the generalization of the definition for a random variable $X \sim P_X$ over $\mathcal{X} = \{x_1, \dots, x_{|\mathcal{X}|}\}$.

Definition 2. *Entropy of $X \sim P_X$, $H(X)$, is the average amount of information by observing the realization of X ,*

$$\begin{aligned} H(X) &\triangleq \sum_{x \in \mathcal{X}} -P_X(x) \log P_X(x) = \mathbb{E}_{P_X}[-\log P_X(X)] \\ &= \sum_{x \in \mathcal{X}} P_X(x) \tilde{H}(X = x). \end{aligned} \quad (2)$$

Remarks:

- 1) Notation: we will use both $H(X)$ and $H(P_X)$
 $H(X)$ depends only on P_X but not on the value of X ; Check that $H(X) = H(2X)$. $H(P_X)$ emphasizes the dependency of H on P_X .
- 2) $\mathbb{E}_{P_X}[F(X)]$ where $F(X) = -\log P_X(X)$ (a function of a random variable X).
- 3) “Average” $\neq$ “Realizations”; When does “average” obtain meaning? when we observe a long seq. of i.i.d (independent and identically distributed) random variables $X_i \sim P_X$; $H(P_X)$ is the **average** amount of information from observing the long sequence of random variables. Information theory (established by Shannon in 1948) highly depends on this “long sequence of i.i.d. random variables,” whose average behavior is predictable by the law of large numbers. These days, there are many attempts to get rid of this assumption and to analyze finite-length behavior.

Example: Binary random variable

Consider a binary random variable $X \sim \text{Bernoulli}(p)$ such that $\Pr(X = 1) = p$ and $\Pr(X = 0) = 1 - p$ for $p \in [0, 1]$.

$$H_B(p) \triangleq H([p, 1 - p]) = -p \log p - (1 - p) \log(1 - p). \quad (3)$$

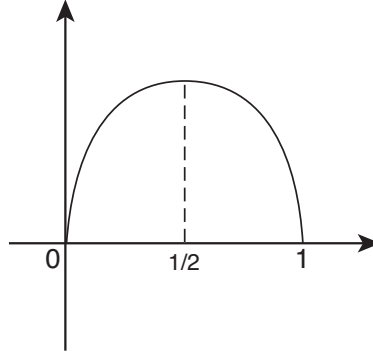


Figure 1: symmetric in p , when $p \approx 0$ or $p \approx 1$ $H_B(p) \approx 0$ (almost deterministic, no information).

Example: Random variable uniformly distributed over a finite set

Consider a random variable U that is uniformly distributed over $\mathcal{U} = \{1, 2, \dots, M\}$, i.e., $\Pr(U = i) = 1/M$ for all $i \in \mathcal{U}$.

$$H(U) = -\sum_{i=1}^M \frac{1}{M} \log \frac{1}{M} = \log M \text{ bits.} \quad (4)$$

Some Properties of $H(\cdot)$:

- 1) $H(P) \geq 0$, with equality iff P is deterministic; ($0 \log 0 = 0$)
- 2) $H(P)$ is continuous in $P \in \mathbb{R}^{\mathcal{X}}$
- 3) H is divisible with successive choices

Example: (H is divisible) Check Fig. 2.

$$H([1/2, 1/3, 1/6]) = H([1/2, 1/2]) + \frac{1}{2}H([2/3, 1/3]) \quad (5)$$

Definition 3 (Joint Entropy). For $(X, Y) \sim P_{XY}$ over $\mathcal{X} \times \mathcal{Y}$,

$$H(X, Y) = H(P_{XY}) = \sum_{(x, y) \in \mathcal{X} \times \mathcal{Y}} P_{XY}(x, y) [-\log P_{XY}(x, y)] \quad (6)$$

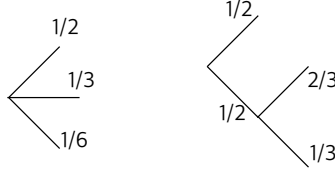


Figure 2: Break down one random variable into sequence of random variables

We next define a notion of conditional entropy, to describe the average amount of information in Y , when the realization of X is observed. Depending on what we mean by “averaging”, there are two sensible definitions.

The conditional entropy of Y , conditioned on $X = x$, is defined as

$$H(Y|X = x) = H(P_{Y|X}(\cdot|x)) \triangleq - \sum_{y \in \mathcal{Y}} P_{Y|X}(y|x) \log P_{Y|X}(y|x). \quad (7)$$

Note that this is a function of x .

We next introduce the second definition of condition entropy, averaged over the joint distribution P_{XY} . This definition is more commonly called *conditional entropy*, which captures the relation between the two random variables.

Definition 4 (Conditional Entropy).

$$\begin{aligned} H(Y|X) &= \mathbb{E}_{P_X}[H(Y|X = x)] \\ &= \sum_{x \in \mathcal{X}} P_X(x) \left(\sum_{y \in \mathcal{Y}} P_{Y|X}(y|x) [-\log P_{Y|X}(y|x)] \right) \\ &= \mathbb{E}_{P_{XY}}[-\log P_{Y|X}(Y|X)] \end{aligned} \quad (8)$$

Example: Consider (X, Y) with the following distribution:

	X=0	X=1
Y=0	1/2	1/3
Y=1	0	1/6

Check that

$$\begin{aligned} H(X) &= H_B(1/2) = 1\text{bit} \\ H(XY) &= H([1/2, 1/3, 1/6]) \\ H(Y) &= H_B(1/6) \\ H(Y|X = 0) &= 0 \\ H(Y|X = 1) &= H_B(1/3) \\ H(Y|X) &= \frac{1}{2} H_B(1/3) \end{aligned}$$

Chain Rule of Entropy:

$$\begin{aligned} H(XY) &= \mathbb{E}_{P_{XY}}[-\log P_{XY}(XY)] \\ &= \mathbb{E}_{P_{XY}}[-\log (P_X(X)P_{Y|X}(Y|X))] \\ &= \mathbb{E}_{P_{XY}}[-\log P_X(X)] + \mathbb{E}_{P_{XY}}[-\log P_{Y|X}(Y|X)] \\ &= H(X) + H(Y|X). \end{aligned} \quad (9)$$

1.2 Mutual Information and Relative Entropy

Definition 5 (Mutual Information).

$$I(X; Y) \triangleq H(Y) - H(Y|X) \quad (10)$$

Remarks:

- 1) $I(X; Y)$ captures the average amount of the decrease of uncertainty (randomness) about Y by observing X ; in other words, it captures the average amount of information about Y contained in X . Here the averaging is over the joint distribution P_{XY} .
- 2) Obviously,

$$X \perp\!\!\!\perp Y \Rightarrow I(X; Y) = 0 \quad (11)$$

It turns out that the reverse is also true, but we will prove it later.

- 3) A trivial relation:

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) = H(Y) - (H(XY) - H(X)) \\ &= H(X) + H(Y) - H(XY) = H(X) - H(X|Y). \end{aligned} \quad (12)$$

Notation: Mutual information is a function of the joint distribution P_{XY} . But it depends on how the joint distribution is represented as a product of P_X and $P_{Y|X}$. Let W denote the conditional distribution $P_{Y|X}$. We often use notation $I(P_X \cdot W)$ to write the mutual information.

Chain Rule of Mutual Information:

$$I(X_1, X_2; Y) = I(X_1; Y) + I(X_2; Y|X_1) \quad (13)$$

Remarks:

- A channel view of the mutual information. Eventually we will make the following statement precise and rigorous. $I(X; Y)$ indicates the amount of information that a terminal who transmits a sequence of symbols X distributed by P_X can send to a receiver that observes a sequence of symbols Y , where the sender and the receiver are connected by a channel with conditional distribution $W = P_{Y|X}$. $I(X_1, X_2; Y)$ means the amount of information that a receiver can obtain by observing Y_i 's where the information is transmitted by two (possibly related) terminals sending X_1 and X_2 . The chain rule says that this amount of information can be broken down into two parts: the first part is about X_1 , and the second part is about X_2 , conditioned on X_1 . Related to this, there is a very important concept called successive cancellation, which we will cover later when we get to the decoding story.

Conditional Mutual Information:

$$I(X; Y|Z) \triangleq H(X|Z) - H(X|Y, Z) = H(Y|Z) - H(Y|X, Z). \quad (14)$$

Note that the conditional mutual information is averaged over Z as

$$I(X; Y|Z) = \sum_z P_Z(z) [H(X|Z = z) - H(X|Y, Z = z)] \quad (15)$$

Remarks:

- In some engineering sense, conditioned on $Z = z$ means that the realization of Z is observed both at the transmitter and the receiver.

Fact 1. Here are some facts that we will prove later.

- $H(X) \leq H(U)$, where U is the uniform distribution over $\mathcal{X}$, and the equality is achieved iff X is uniformly distributed.
- $H(Y) \geq H(Y|X)$, with equality iff $X \perp\!\!\!\perp Y$.
- $I(X; Y) \geq 0$, with equality iff $X \perp\!\!\!\perp Y$.

To prove these facts, we need to define another (possibly the most important) information measure, relative entropy.

Definition 6. The **relative entropy**, or **Kullback-Leibler Divergence** between two distributions, P and Q , on the same alphabet $\mathcal{X}$, is

$$D(P\|Q) \triangleq \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} = \mathbb{E}_{P_X} \left[\log \frac{P(X)}{Q(X)} \right]. \quad (16)$$

Remarks:

- 1) $D(P\|Q)$ is in some sense a measure of the distance between two distributions, P and Q .
- 2) When $P = Q$, the relative entropy $D(P\|Q) = 0$.
- 3) $D(P\|Q)$ often appears in error analysis: Suppose $X_1, \dots, X_n$ are i.i.d. from distribution Q . We know that if n is large enough, the empirical distribution of the string would be close to Q . A more precise statement is that the probability that the empirical distribution is some $P \neq Q$ is small. In fact, it is exponentially small and

$$\Pr(*) \approx \exp[-nD(P\|Q)]. \quad (17)$$

We will make this statement more precise and prove the statement in later lectures.

- 4) But the relative entropy is different from other classical distances such as Euclidean distance in the sense that the relative entropy does not hold symmetry. Note that $D(P\|Q) \neq D(Q\|P)$ for general P and Q .

Fact 2. Here are some easy facts:

$$H(P_X) = H(P_U) - D(P_X\|P_U) \quad (18)$$

$$I(P_X \cdot W) = D(P_{XY}\|[P_X \cdot P_Y]), \quad (19)$$

where P_U is the uniform distribution over $\mathcal{X}$ and $[P_X \cdot P_Y]$ is the product of marginal distributions of X and Y .

Proof of Eq. (18). Assume that $X, U \in \mathcal{X} = \{1, 2, \dots, M\}$.

$$\begin{aligned} D(P_X\|P_U) &= \sum_{x \in \mathcal{X}} P_X(x) \log \frac{P_X(x)}{P_U(x)} \\ &= \sum_{x \in \mathcal{X}} P_X(x) \log \frac{P_X(x)}{1/M} \\ &= \log M - H(P_X) = H(P_U) - H(P_X). \end{aligned} \quad (20)$$

□

Proof of Eq. (19).

$$\begin{aligned}
D(P_{XY} \| [P_X \cdot P_Y]) &= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} P_{XY}(x,y) \log \frac{P_{XY}(x,y)}{P_X(x)P_Y(y)} \\
&= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} P_{XY}(x,y) \log \frac{P_{Y|X}(y|x)}{P_Y(y)} \\
&= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} P_{XY}(x,y) \log P_{Y|X}(y|x) - \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} P_{XY}(x,y) \log P_Y(y) \\
&= -H(Y|X) + H(Y) = I(X;Y) = I(P_X \cdot P_Y).
\end{aligned} \tag{21}$$

□

1.3 Some Inequalities about Information Quantities

While information theory is all about inequalities, here is the first important one.

Theorem 1 (Information Inequality). $D(P \| Q) \geq 0$ with equality iff $P = Q$.

Lemma 1. For any $u > 0$ and $t > -1$,

$$\begin{aligned}
\log u &\leq u - 1, \\
\log(1 + t) &\leq t.
\end{aligned} \tag{22}$$

Proof of Theorem 1.

$$\begin{aligned}
-D(P \| Q) &= \sum_x P(x) \log \frac{Q(x)}{P(x)} \\
&\leq \sum_x P(x) \left[\frac{Q(x)}{P(x)} - 1 \right] = 0
\end{aligned} \tag{23}$$

□

By using Theorem 1, we can easily prove Fact. 1 in p.4.

Corollary 1. $H(P) = H(U) - D(P \| U)$, hence $H(P) \leq H(U)$ with equality only if $P = U$.

Uniform distribution maximizes entropy.

Corollary 2. $I(X;Y) = I(P_X \cdot P_Y) = D(P_{XY} \| [P_X \cdot P_Y]) \geq 0$ with equality iff $X \perp\!\!\!\perp Y$.

Obviously, this implies $H(Y) \geq H(Y|X)$, which means conditioning always decreases the entropy on average.

Corollary 3 (Log-sum Inequality). Given $a_i, b_i \geq 0$, $i = 1, 2, \dots, n$, let $a = \sum_{i=1}^n a_i$ and $b = \sum_{i=1}^n b_i$. Then we have

$$\sum_i a_i \log \frac{a_i}{b_i} \geq a \log \frac{a}{b}, \tag{24}$$

with equality iff $\frac{a_i}{a} = \frac{b_i}{b}$, $\forall i$.

Proof. Consider $k_a > 0$ and $k_b > 0$ that make $k_a a = \sum_i k_a a_i = 1$ and $k_b b = \sum_i k_b b_i = 1$. Note that the inequality in (24) is equivalent to

$$\sum_i (k_a a_i) \log \frac{(k_a a_i)}{(k_b b_i)} \geq (k_a a) \log \frac{(k_a a)}{(k_b b)}. \tag{25}$$

This goes back to the information inequality in Theorem 1. □

* * *

We next discuss some convexity properties of H, I and D .

Theorem 2. $D(P\|Q)$ is convex in the pair (P, Q) ; that is, for any distribution pairs (P_0, Q_0) and (P_1, Q_1) , when we define $P_\theta = (1 - \theta)P_0 + \theta P_1$ and $Q_\theta = (1 - \theta)Q_0 + \theta Q_1$, we have

$$D(P_\theta\|Q_\theta) \leq (1 - \theta)D(P_0\|Q_0) + \theta D(P_1\|Q_1). \quad (26)$$

Proof. For every x ,

$$P_\theta(x) \log \frac{P_\theta(x)}{Q_\theta(x)} \leq (1 - \theta)P_0(x) \log \frac{P_0(x)}{Q_0(x)} + \theta P_1(x) \log \frac{P_1(x)}{Q_1(x)} \quad (27)$$

by log-sum inequality. By summing up over $x \in \mathcal{X}$, we obtain (26). $\square$

Corollary 4. $H(P)$ is a concave function of P . That is

$$H(P_\theta) \geq (1 - \theta)H(P_0) + \theta H(P_1). \quad (28)$$

Example: Recall the binary entropy curve.

Proof. Quick proof: $H(P) = H(U) - D(P\|U)$, where U is the uniform distribution over $\mathcal{X}$. The concavity of H is then follows from the convexity of D .

Alternative (more interesting) proof: Consider two random variables $X_0 \sim P_0$ and $X_1 \sim P_1$. Let there be a switch random variable Z , independent of X_0 and X_1 , that is Bernoulli distributed with $\Pr(Z = 0) = 1 - \theta$ and $\Pr(Z = 1) = \theta$ as in Fig. 3. Define a new random variable X such that $X = X_i$ for $Z = i$, $i = 0, 1$. Then we have

$$\begin{aligned} H(X) &= H(P_\theta) \geq H(X|Z) \quad \text{conditioning reduces the entropy} \\ &= (1 - \theta)H(X|Z = 0) + \theta H(X|Z = 1) = (1 - \theta)H(P_0) + \theta H(P_1). \end{aligned} \quad (29)$$

$\square$

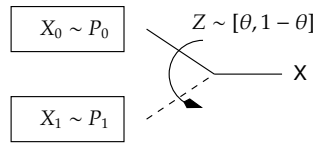


Figure 3: $H(P)$ is concave in P

Remarks: In a few lectures later (possibly next week), we will eventually show that $H(X)$ is the rate at which a source X can be encoded (compressed). Corollary 4 tells us that if we know Z , i.e., how X is mixed from two distributions, we need a lower rate to encode the source.

Corollary 5. $I(P_X \cdot W)$ is concave in P_X for any fixed W .

Proof. Consider the source we considered in the proof of the previous corollary. We feed the source to a channel W as in Fig. 4. What we will show is

$$I(P_\theta \cdot W) \geq (1 - \theta)I(P_0 \cdot W) + \theta I(P_1 \cdot W). \quad (30)$$

To show this, consider the following channel rule.

$$\begin{aligned} I(X, Z; Y) &= I(X; Y) + \cancel{I(Z; Y|X)}^0 \\ &= I(Z; Y) + I(X; Y|Z). \end{aligned} \quad (31)$$

Note that Z is used to construct X , and is independent of Y given X , and thus $I(Z; Y|X) = 0$. So we have

$$\begin{aligned} I(P_\theta \cdot W) &= I(X; Y) = I(Z; Y) + I(X; Y|Z) \\ &\geq I(X; Y|Z) \\ &= \Pr(Z = 0)I(X; Y|Z = 0) + \Pr(Z = 1)I(X; Y|Z = 1) \\ &= (1 - \theta)I(P_0 \cdot W) + \theta I(P_1 \cdot W). \end{aligned} \quad (32)$$

□

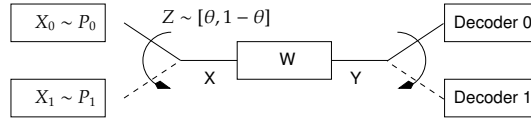


Figure 4: $I(P_X \cdot W)$ is concave in P_X

Remarks: We will show much later in the course that the mutual information $I(P_X \cdot W)$ is related to the amount of information, in bits, that one can transmit over a channel W , with an input of distribution P_X . In this sense, the corollary says that to transmit more bits through a given channel, one should randomize the input. If a part of randomness in X , namely Z , is revealed to the receiver, this takes away some information contents in the source, and reduces the amount of information one could have transmitted over the channel.

Corollary 6. $I(P_X \cdot W)$ is convex in W for any fixed P_X .

Proof. Consider two channels W_0 and W_1 , and an independent switch random variable Z , with $\Pr(Z = 0) = 1 - \theta$ and $\Pr(Z = 1) = \theta$, as in Fig. 5. A new channel W is formed by $W = W_i$ from $Z = i$, $i = 0, 1$. By chain rules,

$$\begin{aligned} I(X; Y, Z) &= I(X; Y|Z) + \cancel{I(X; Z)}^0 \\ &= I(X; Z|Y) + I(X; Y), \end{aligned} \quad (33)$$

where $I(X; Z) = 0$ since $X \perp\!\!\!\perp Z$. Thus we have

$$I(P_X \cdot W_\theta) = I(X; Y) \leq I(X; Y|Z) = (1 - \theta)I(P_X \cdot W_0) + \theta I(P_X \cdot W_1). \quad (34)$$

□

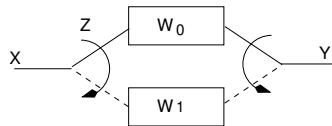


Figure 5: $I(P_X \cdot W)$ is concave in P_X

Remarks: This result says that less bits can be communicated when there is a randomization over the channel, not known at the transmitter and the receiver. This is quite intuitive.

1.4 Data Processing Inequality & Fano's Inequality

There are two more inequalities, which are in some sense more important than the previous ones as they are certainly used more often.

Definition 7. Random variables, X, Y, Z are said to form a Markov chain in that order (denoted by $X \rightarrow Y \rightarrow Z$) iff

$$P_{XYZ}(x, y, z) = P_X(x)P_{Y|X}(y|x)P_{Z|YX}(z|yx) = P_X(x)P_{Y|X}(y|x)P_{Z|Y}(z|y), \forall x, y, z. \quad (35)$$

Remarks:

- 1) The above relation says $P_{Z|YX} = P_{Z|Y}$, or in words, given Y , the two random variables X and Z are independent. This is equivalent to $P_{XZ|Y} = P_{X|Y}P_{Z|Y}$.
- 2) Clearly, $X \rightarrow Y \rightarrow Z$ is equivalent to $Z \rightarrow Y \rightarrow X$.

Theorem 3 (Data Processing Inequality). If $X \rightarrow Y \rightarrow Z$,

$$I(X; Y) \geq I(X; Z) \quad (36)$$

with equality iff $I(X; Y|Z) = 0$, or equivalently, $X \rightarrow Z \rightarrow Y$.

Proof.

$$\begin{aligned} I(X; Y, Z) &= I(X; Y) + \overbrace{I(X; Z|Y)}^{\rightarrow 0} \\ &= I(X; Z) + I(X; Y|Z). \end{aligned} \quad (37)$$

□

Discussion: Consider the case when $I(X; Y) = I(X; Z)$, i.e. when $I(X; Y|Z) = 0$. For this case, all the information about X one can obtain by observing Y can also be obtained by observing Z . Thus the observation of Z is sufficient to infer any information about X that can be inferred by observing Y . In detection and estimation, we have a similar notion called *sufficient statistics*. One say that Z , which is a deterministic function of Y , say $Z(Y)$, is a sufficient statistic iff the likelihood ratio can be written as a function of Z . That is $P_{Y|X}(y|x_1)/P_{Y|X}(y|x_2) = F(Z, x_1, x_2)$, and does not depend on y .

* * *

Finally, we next learn about Fano's inequality. Consider $(X, Y) \sim P_{XY}$. Denote the alphabet of X as $\mathcal{X}$. Suppose we observe Y and try to make a guess on X . To do so, we form a function $\hat{X}(Y)$. Let $\mathbb{1}_E$ be the indicator of the event $\hat{X} = X$. We often refer to the probability $\Pr(\mathbb{1}_E = 0)$ as the probability of error P_e . Fano's inequality says that *the probability of error cannot be too small if $H(X|Y)$ is large, i.e., the correct guess of X happens only when the residual randomness of X given Y is small.*

Theorem 4 (Fano's inequality).

$$H(X|Y) \leq \log 2 + P_e \cdot \log(|\mathcal{X}| - 1) \quad (38)$$

Proof.

$$\begin{aligned}
& H(X, \mathbb{1}_E | Y) \\
&= H(X | Y) + \cancel{H(\mathbb{1}_E | X, Y)} \xrightarrow{0} \quad \mathbb{1}_E \text{ is determined by } X, Y \\
&= H(\mathbb{1}_E | Y) + H(X | \mathbb{1}_E, Y) \\
&= H(\mathbb{1}_E | Y) + \Pr(\mathbb{1}_E = 0)H(X | \mathbb{1}_E = 0, Y) + \Pr(\mathbb{1}_E = 1)H(X | \mathbb{1}_E = 1, Y) \xrightarrow{0} \\
&\leq \log 2 + P_e H(X | \mathbb{1}_E = 0, Y) \\
&\leq \log 2 + P_e \cdot \log(|\mathcal{X}| - 1).
\end{aligned} \tag{39}$$

□

Remarks: Fano's inequality is one of the most important inequalities in information theory, although it is very simple. Most of information theory problems seek to answer the question of “what is the fundamental limits on achievable performance (in compressions, communications or data processing) that still guarantees reliable guessing (decoding, decision, etc.)?” Thus, to claim that some limit is the best possible on the system, we need an inequality like Fano's inequality (to prove the converse result) that tells “we cannot have a correct guessing when we go beyond this limit.”